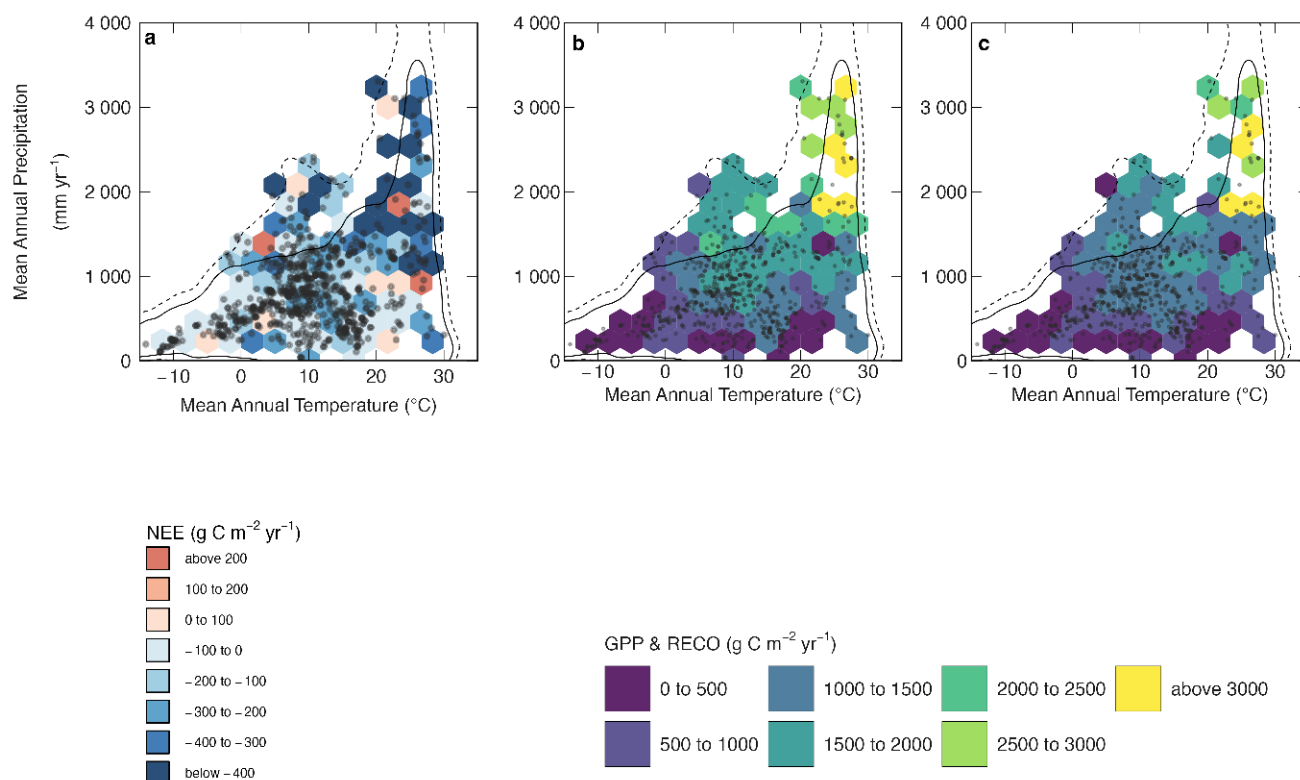


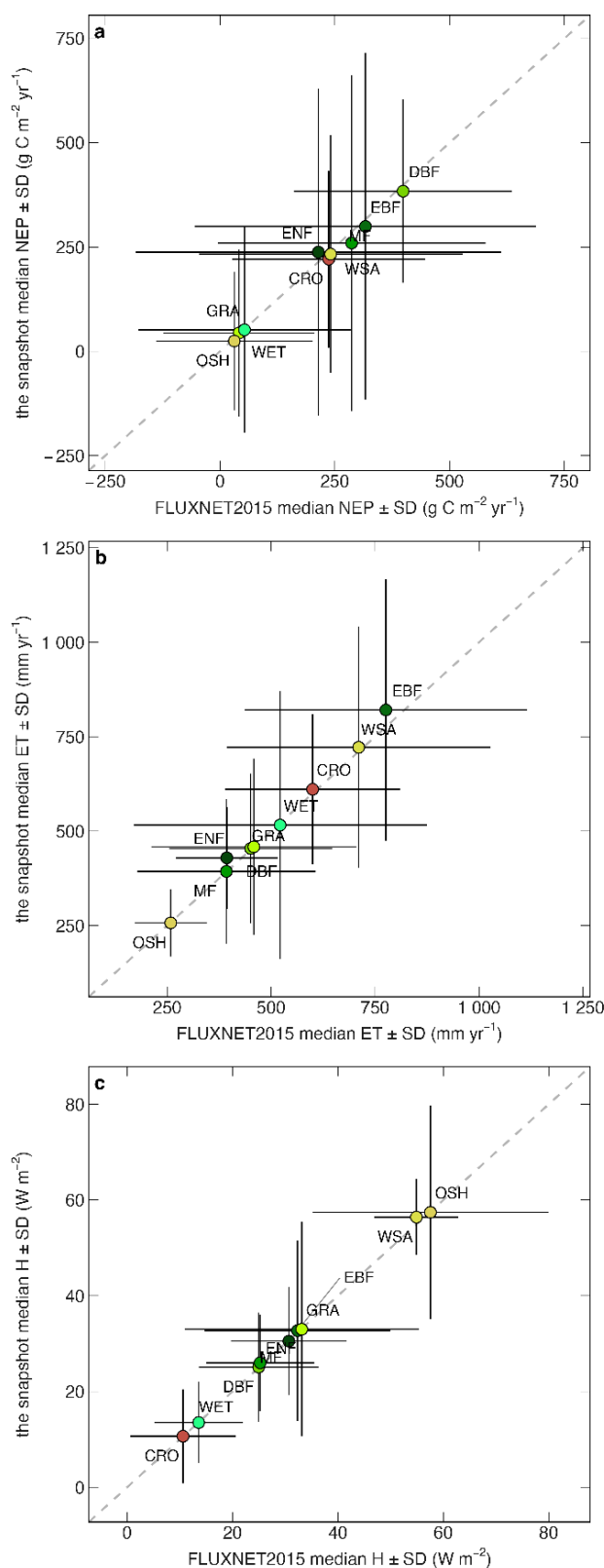
Supplementary Figure S1

Whittaker climate–space distribution of the snapshot network's net ecosystem exchange (NEE), gross primary productivity (GPP) and ecosystem respiration (RECO). Towers are placed in mean–annual–temperature by mean–annual–precipitation space (WorldClim v2.1) and coloured by the hexagon–binned median of each site's flux. Solid and dashed black lines show the 95% and 99% highest–density regions of global ice–free land in the same climate space, for comparison with the main text's equivalent NEE distribution. Data used in the analyses are described in Section 2.5.



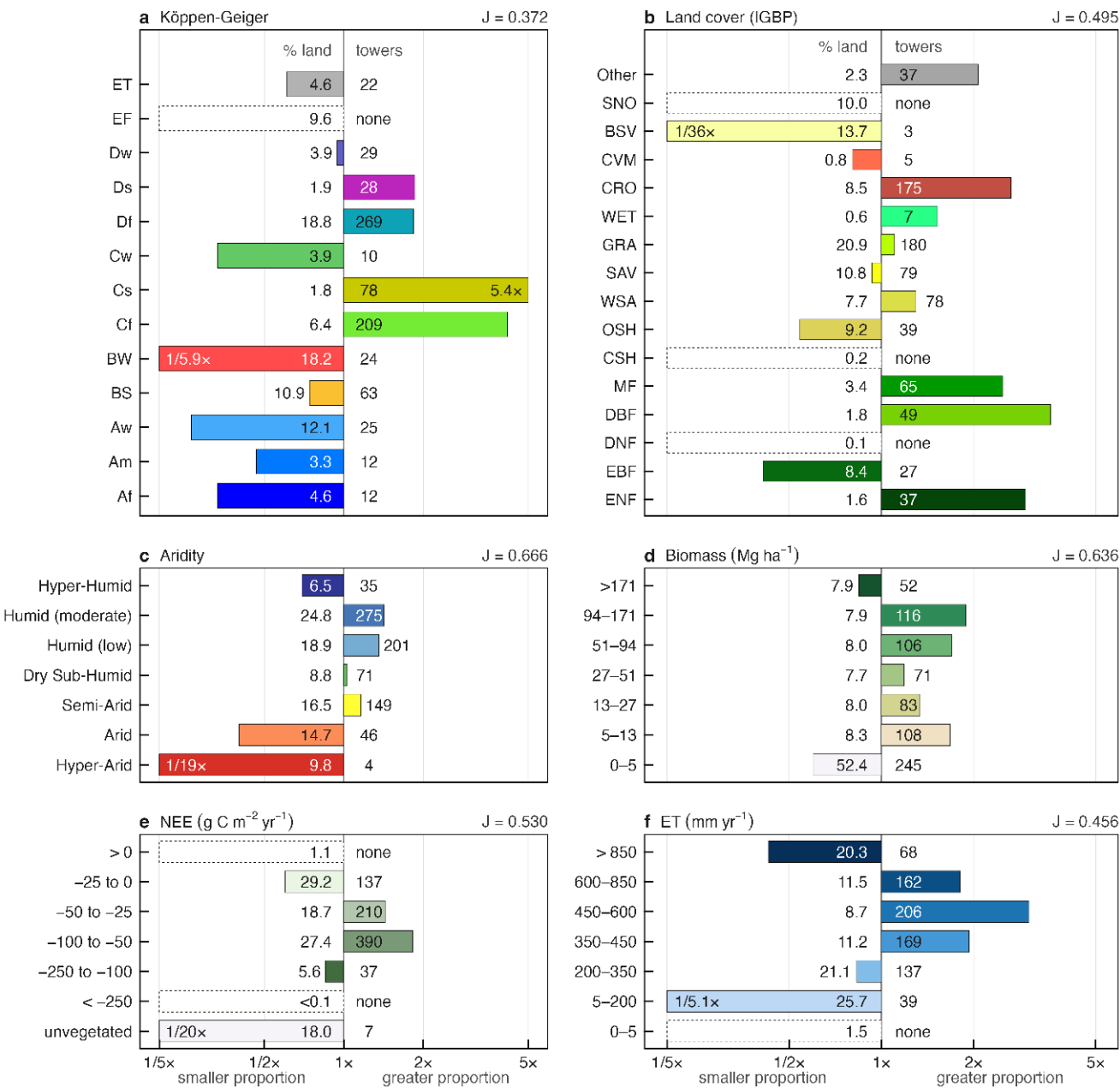
Supplementary Figure S2

Comparison of per-IGBP-class median net ecosystem production (NEP), evapotranspiration (ET) and sensible heat flux (H) between the FLUXNET2015 release and the snapshot, restricted to site-years present and qualifying in both datasets for a given site and calendar year. This matched comparison isolates differences arising from updated flux-processing software from differences arising from network composition change, complementing the full-network comparison in the main text. Data used in the analyses are described in Section 2.5.



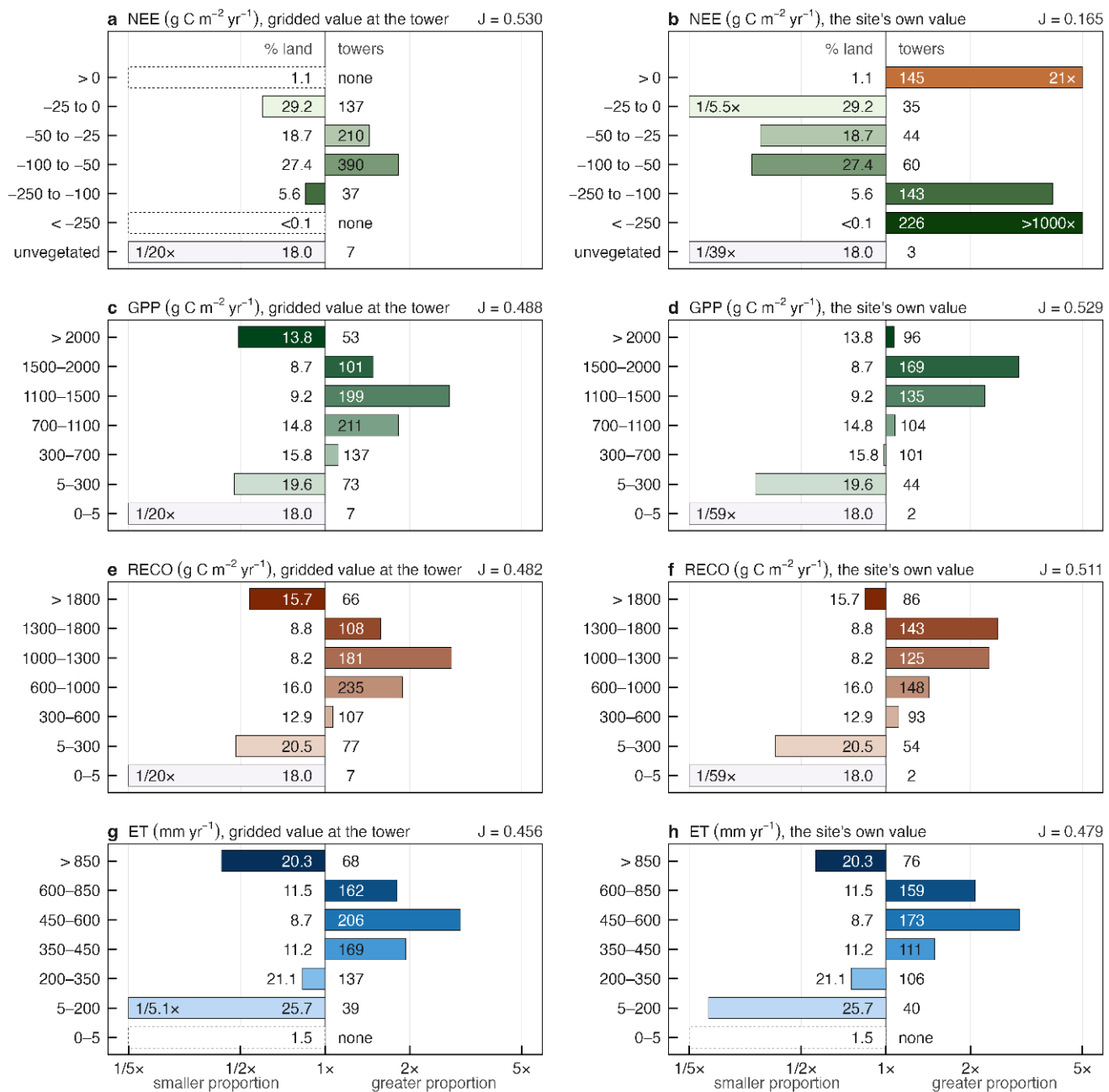
Supplementary Figure S3

Network sampling of the snapshot (n=781 sites) across six axes --- Köppen–Geiger climate class, land cover (IGBP), aridity, aboveground biomass, net ecosystem exchange and evapotranspiration --- using the gridded value at the tower coordinate rather than each site's own measurement. This is the companion to the main text's sampling figure, which instead uses the site's own value; the two together show how much the network's apparent sampling depends on that choice. Network sampling analysis is described in Section 2.7.



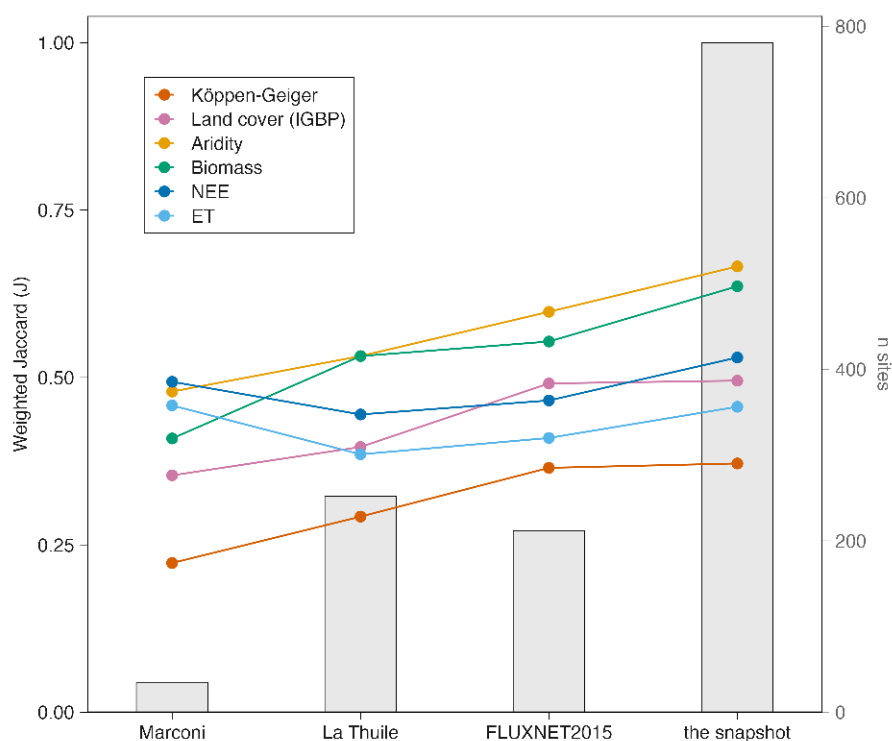
Supplementary Figure S4

Network sampling of the snapshot for four fluxes -- net ecosystem exchange, gross primary productivity, ecosystem respiration and evapotranspiration -- each compared using both the gridded value at the tower and the site's own value. Extends the main text's sampling figure (which covers net ecosystem exchange and evapotranspiration) to add gross primary productivity and ecosystem respiration. Network sampling analysis is described in Section 2.7.



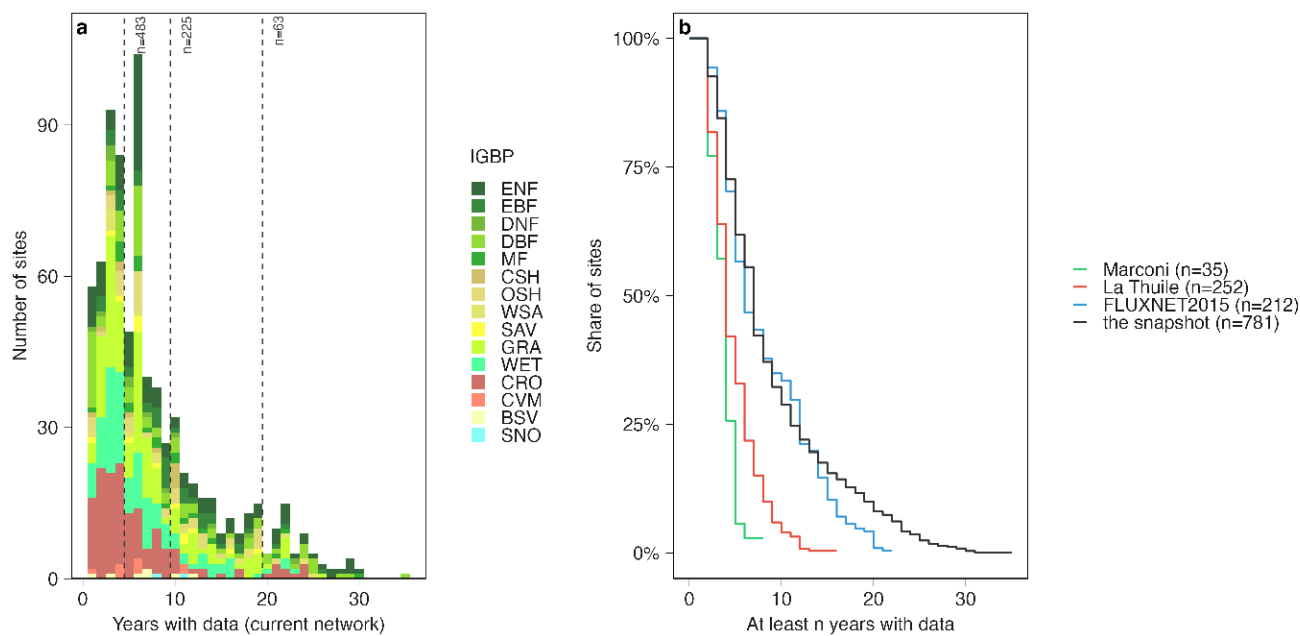
Supplementary Figure S5

Network sampling (gridded value at the tower) tracked across four generations of FLUXNET data collection -- Marconi, La Thuile, FLUXNET2015 and the snapshot -- for the same six axes as the main text's sampling figure. Shows how the network's coverage of climate, land-cover, aridity, biomass and flux space has changed as the network has grown. Network sampling analysis is described in Section 2.7.



Supplementary Figure S6

Record length of the snapshot and of three prior FLUXNET data collections (Marconi, La Thuile, FLUXNET2015). Panel a shows the distribution of years with data per site in the snapshot, by IGBP class. Panel b shows, for each collection, the share of its sites with at least a given number of years of data. Data used in the analyses are described in Section 2.5.



Supplementary Table S1. Regional networks contributing to the snapshot: network code, name, processing hub, number of sites and number of site-years.

network_code	network_name	processing_hub	n_sites	n_site_years
AMF	AmeriFlux	AmeriFlux	381	2891
EUF	European Fluxes Database	ICOS	138	959
ICOS	ICOS	ICOS	80	983
JPF	JapanFlux	ICOS	54	507
TERN	TERN	TERN	52	430
CNF	ChinaFLUX	ICOS	32	246
KOF	KoFlux	ICOS	21	82
FLX	not affiliated with a regional network	ICOS	18	87
SAEON	South African Environmental Observation Network	ICOS	5	15

Supplementary Table S2. Record length by IGBP class: sites, and sites with at least 5, 10 and 20 years of data (overall and for a qualifying annual NEE).

igbp_class	n_sites	data_ge5	nee_ge5	data_ge10	nee_ge10	data_ge20	nee_ge20
ENF	114	88	69	46	36	19	12
EBF	44	34	23	17	10	3	2
DNF	13	8	3	4	3	1	1
DBF	81	53	41	30	25	13	9
MF	24	18	9	9	4	3	2
CSH	12	7	6	5	0	0	0
OSH	41	30	20	16	9	1	0
WSA	18	12	11	8	6	3	3
SAV	14	11	8	6	3	0	0
GRA	146	86	58	35	25	6	3
WET	117	61	32	23	14	4	2
CRO	139	63	41	21	16	10	6
CVM	9	5	3	2	0	0	0
BSV	7	5	2	2	0	0	0
SNO	2	2	0	1	0	0	0
Total	781	483	326	225	151	63	40

Supplementary Table S3. Network sampling ratios by axis and class: land share, tower share and the log2 sampling ratio, for both the gridded value at the tower and the site's own value.

axis	comparison	class	land_share	tower_share	tower_count	n_towers_classified	sampling_ratio	log2_ratio	note
koppen	geo_vs_geo	Af	0.04610	0.01540	12	781	0.3330	-1.5900	NA
koppen	geo_vs_geo	Am	0.03300	0.01540	12	781	0.4660	-1.1000	NA
koppen	geo_vs_geo	Aw	0.12100	0.03200	25	781	0.2650	-1.9200	NA
koppen	geo_vs_geo	BS	0.10900	0.08070	63	781	0.7420	-0.4310	NA
koppen	geo_vs_geo	BW	0.18200	0.03070	24	781	0.1690	-2.5600	NA
koppen	geo_vs_geo	Cf	0.06430	0.26800	209	781	4.1600	2.0600	NA
koppen	geo_vs_geo	Cs	0.01840	0.09990	78	781	5.4300	2.4400	NA
koppen	geo_vs_geo	Cw	0.03850	0.01280	10	781	0.3320	-1.5900	NA
koppen	geo_vs_geo	Df	0.18800	0.34400	269	781	1.8400	0.8770	NA
koppen	geo_vs_geo	Ds	0.01930	0.03590	28	781	1.8600	0.8930	NA
koppen	geo_vs_geo	Dw	0.03940	0.03710	29	781	0.9440	-0.0837	NA
koppen	geo_vs_geo	EF	0.09560	0.00000	0	781	0.0000	NA	NA
koppen	geo_vs_geo	ET	0.04640	0.02820	22	781	0.6070	-0.7200	NA
koppen	geo_vs_data	Af	0.04610	0.02000	15	750	0.4340	-1.2100	NA
koppen	geo_vs_data	Am	0.03300	0.01330	10	750	0.4040	-1.3100	NA
koppen	geo_vs_data	Aw	0.12100	0.03330	25	750	0.2750	-1.8600	NA
koppen	geo_vs_data	BS	0.10900	0.10300	77	750	0.9440	-0.0826	NA
koppen	geo_vs_data	BW	0.18200	0.02670	20	750	0.1470	-2.7700	NA
koppen	geo_vs_data	Cf	0.06430	0.28000	210	750	4.3500	2.1200	NA
koppen	geo_vs_data	Cs	0.01840	0.09870	74	750	5.3600	2.4200	NA
koppen	geo_vs_data	Cw	0.03850	0.01070	8	750	0.2770	-1.8500	NA
koppen	geo_vs_data	Df	0.18800	0.31300	235	750	1.6700	0.7400	NA
koppen	geo_vs_data	Ds	0.01930	0.02670	20	750	1.3800	0.4660	NA
koppen	geo_vs_data	Dw	0.03940	0.04000	30	750	1.0200	0.0236	NA
koppen	geo_vs_data	EF	0.09560	0.00000	0	750	0.0000	NA	NA
koppen	geo_vs_data	ET	0.04640	0.03470	26	750	0.7470	-0.4210	NA
igbp	geo_vs_geo	BSV	0.13700	0.00384	3	781	0.0279	-5.1600	NA
igbp	geo_vs_geo	CRO	0.08460	0.22400	175	781	2.6500	1.4100	NA
igbp	geo_vs_geo	CSH	0.00238	0.00000	0	781	0.0000	NA	NA
igbp	geo_vs_geo	CVM	0.00795	0.00640	5	781	0.8060	-0.3120	NA
igbp	geo_vs_geo	DBF	0.01760	0.06270	49	781	3.5700	1.8300	NA
igbp	geo_vs_geo	DNF	0.00114	0.00000	0	781	0.0000	NA	NA
igbp	geo_vs_geo	EBF	0.08410	0.03460	27	781	0.4110	-1.2800	NA
igbp	geo_vs_geo	ENF	0.01610	0.04740	37	781	2.9500	1.5600	NA
igbp	geo_vs_geo	GRA	0.20900	0.23000	180	781	1.1000	0.1420	NA

Supplementary Table S3. Network sampling ratios by axis and class: land share, tower share and the log2 sampling ratio, for both the gridded value at the tower and the site's own value. (continued)

axis	comparison	class	land_share	tower_share	tower_count	n_towers_classified	sampling_ratio	log2_ratio	note
igbp	geo_vs_geo	MF	0.03350	0.08320	65	781	2.4800	1.3100	NA
igbp	geo_vs_geo	OSH	0.09210	0.04990	39	781	0.5420	-0.8840	NA
igbp	geo_vs_geo	Other	0.02290	0.04740	37	781	2.0700	1.0500	NA
igbp	geo_vs_geo	SAV	0.10800	0.10100	79	781	0.9330	-0.1010	NA
igbp	geo_vs_geo	SNO	0.09980	0.00000	0	781	0.0000	NA	NA
igbp	geo_vs_geo	WET	0.00589	0.00896	7	781	1.5200	0.6060	NA
igbp	geo_vs_geo	WSA	0.07700	0.09990	78	781	1.3000	0.3750	NA
igbp	geo_vs_data	BSV	0.13700	0.00896	7	781	0.0652	-3.9400	NA
igbp	geo_vs_data	CRO	0.08460	0.17800	139	781	2.1000	1.0700	NA
igbp	geo_vs_data	CSH	0.00238	0.01540	12	781	6.4600	2.6900	NA
igbp	geo_vs_data	CVM	0.00795	0.01150	9	781	1.4500	0.5360	NA
igbp	geo_vs_data	DBF	0.01760	0.10400	81	781	5.9000	2.5600	NA
igbp	geo_vs_data	DNF	0.00114	0.01660	13	781	14.6000	3.8600	NA
igbp	geo_vs_data	EBF	0.08410	0.05630	44	781	0.6700	-0.5780	NA
igbp	geo_vs_data	ENF	0.01610	0.14600	114	781	9.0900	3.1800	NA
igbp	geo_vs_data	GRA	0.20900	0.18700	146	781	0.8950	-0.1600	NA
igbp	geo_vs_data	MF	0.03350	0.03070	24	781	0.9170	-0.1250	NA
igbp	geo_vs_data	OSH	0.09210	0.05250	41	781	0.5700	-0.8120	NA
igbp	geo_vs_data	Other	0.02290	0.00000	0	781	0.0000	NA	NA
igbp	geo_vs_data	SAV	0.10800	0.01790	14	781	0.1650	-2.6000	NA
igbp	geo_vs_data	SNO	0.09980	0.00256	2	781	0.0257	-5.2800	NA
igbp	geo_vs_data	WET	0.00589	0.15000	117	781	25.4000	4.6700	NA
igbp	geo_vs_data	WSA	0.07700	0.02300	18	781	0.2990	-1.7400	NA
aridity	geo_vs_geo	Arid	0.14700	0.05890	46	781	0.4010	-1.3200	NA
aridity	geo_vs_geo	Dry Sub-Humid	0.08830	0.09090	71	781	1.0300	0.0418	NA
aridity	geo_vs_geo	Humid (low)	0.18900	0.25700	201	781	1.3600	0.4420	NA
aridity	geo_vs_geo	Humid (moderate)	0.24800	0.35200	275	781	1.4200	0.5040	NA
aridity	geo_vs_geo	Hyper-Arid	0.09810	0.00512	4	781	0.0522	-4.2600	NA
aridity	geo_vs_geo	Hyper-Humid	0.06450	0.04480	35	781	0.6950	-0.5250	NA
aridity	geo_vs_geo	Semi-Arid	0.16500	0.19100	149	781	1.1600	0.2130	NA
aridity	geo_vs_data	Arid	0.14700	0.05050	29	574	0.3440	-1.5400	NA
aridity	geo_vs_data	Dry Sub-Humid	0.08830	0.09580	55	574	1.0900	0.1180	NA
aridity	geo_vs_data	Humid (low)	0.18900	0.20200	116	574	1.0700	0.0933	NA
aridity	geo_vs_data	Humid (moderate)	0.24800	0.37800	217	574	1.5200	0.6060	NA
aridity	geo_vs_data	Hyper-Arid	0.09810	0.00000	0	574	0.0000	NA	NA

Supplementary Table S3. Network sampling ratios by axis and class: land share, tower share and the log2 sampling ratio, for both the gridded value at the tower and the site's own value. (continued)

axis	comparison	class	land_share	tower_share	tower_count	n_towers_classified	sampling_ratio	log2_ratio	note
aridity	geo_vs_data	Hyper-Humid	0.064500	0.08540	49	574	1.32e+00	0.404	NA
aridity	geo_vs_data	Semi-Arid	0.165000	0.18800	108	574	1.14e+00	0.193	NA
biomass	geo_vs_geo	1	0.524000	0.31400	245	781	5.99e-01	-0.739	NA
biomass	geo_vs_geo	2	0.082500	0.13800	108	781	1.68e+00	0.745	NA
biomass	geo_vs_geo	3	0.079500	0.10600	83	781	1.34e+00	0.418	NA
biomass	geo_vs_geo	4	0.076600	0.09090	71	781	1.19e+00	0.248	NA
biomass	geo_vs_geo	5	0.080100	0.13600	106	781	1.69e+00	0.761	NA
biomass	geo_vs_geo	6	0.078700	0.14900	116	781	1.89e+00	0.916	NA
biomass	geo_vs_geo	7	0.078900	0.06660	52	781	8.44e-01	-0.244	NA
biomass	geo_vs_data	1	0.524000	0.31400	245	781	5.99e-01	-0.739	NA
biomass	geo_vs_data	2	0.082500	0.13800	108	781	1.68e+00	0.745	NA
biomass	geo_vs_data	3	0.079500	0.10600	83	781	1.34e+00	0.418	NA
biomass	geo_vs_data	4	0.076600	0.09090	71	781	1.19e+00	0.248	NA
biomass	geo_vs_data	5	0.080100	0.13600	106	781	1.69e+00	0.761	NA
biomass	geo_vs_data	6	0.078700	0.14900	116	781	1.89e+00	0.916	NA
biomass	geo_vs_data	7	0.078900	0.06660	52	781	8.44e-01	-0.244	NA
nee	geo_vs_geo	1	0.180000	0.00896	7	781	4.98e-02	-4.330	NA
nee	geo_vs_geo	2	0.000144	0.00000	0	781	0.00e+00	NA	NA
nee	geo_vs_geo	3	0.055600	0.04740	37	781	8.52e-01	-0.231	NA
nee	geo_vs_geo	4	0.274000	0.49900	390	781	1.82e+00	0.868	NA
nee	geo_vs_geo	5	0.187000	0.26900	210	781	1.44e+00	0.522	NA
nee	geo_vs_geo	6	0.292000	0.17500	137	781	6.00e-01	-0.737	NA
nee	geo_vs_geo	7	0.010800	0.00000	0	781	0.00e+00	NA	NA
nee	geo_vs_data	1	0.180000	0.00457	3	656	2.54e-02	-5.300	NA
nee	geo_vs_data	2	0.000144	0.34500	226	656	2.39e+03	11.200	NA
nee	geo_vs_data	3	0.055600	0.21800	143	656	3.92e+00	1.970	NA
nee	geo_vs_data	4	0.274000	0.09150	60	656	3.34e-01	-1.580	NA
nee	geo_vs_data	5	0.187000	0.06710	44	656	3.58e-01	-1.480	NA
nee	geo_vs_data	6	0.292000	0.05340	35	656	1.82e-01	-2.450	NA
nee	geo_vs_data	7	0.010800	0.22100	145	656	2.05e+01	4.360	NA
et	geo_vs_geo	1	0.014700	0.00000	0	781	0.00e+00	NA	NA
et	geo_vs_geo	2	0.257000	0.04990	39	781	1.94e-01	-2.360	NA
et	geo_vs_geo	3	0.211000	0.17500	137	781	8.31e-01	-0.268	NA
et	geo_vs_geo	4	0.112000	0.21600	169	781	1.93e+00	0.950	NA
et	geo_vs_geo	5	0.087200	0.26400	206	781	3.03e+00	1.600	NA

Supplementary Table S3. Network sampling ratios by axis and class: land share, tower share and the log2 sampling ratio, for both the gridded value at the tower and the site's own value. (continued)

axis	comparison	class	land_share	tower_share	tower_count	n_towers_classified	sampling_ratio	log2_ratio	note
et	geo_vs_geo	6	0.1150	0.2070	162	781	1.800	0.850	NA
et	geo_vs_geo	7	0.2030	0.0871	68	781	0.429	-1.220	NA
et	geo_vs_data	1	0.0147	0.0000	0	665	0.000	NA	NA
et	geo_vs_data	2	0.2570	0.0602	40	665	0.234	-2.090	NA
et	geo_vs_data	3	0.2110	0.1590	106	665	0.755	-0.406	NA
et	geo_vs_data	4	0.1120	0.1670	111	665	1.490	0.576	NA
et	geo_vs_data	5	0.0872	0.2600	173	665	2.980	1.580	NA
et	geo_vs_data	6	0.1150	0.2390	159	665	2.080	1.060	NA
et	geo_vs_data	7	0.2030	0.1140	76	665	0.563	-0.830	NA

Supplementary Table S4. Bowen ratio (sensible / latent heat flux) by IGBP class: median and interquartile range.

igbp_class	n_sites	n_site_years	bowen_median	bowen_p25	bowen_p75	flag_small_n
ENF	103	963	0.890	0.542	1.130	FALSE
EBF	40	282	0.417	0.250	0.858	FALSE
DNF	7	63	1.460	0.815	1.510	FALSE
DBF	64	593	0.560	0.418	0.776	FALSE
MF	23	150	0.629	0.407	0.910	FALSE
CSH	11	57	0.738	0.347	1.140	FALSE
OSH	34	239	1.210	0.458	3.160	FALSE
WSA	17	155	1.220	0.733	1.920	FALSE
SAV	13	88	1.350	1.050	2.100	FALSE
GRA	130	761	0.463	0.190	0.826	FALSE
WET	95	479	0.357	0.176	0.615	FALSE
CRO	113	598	0.327	0.208	0.439	FALSE
CVM	7	26	0.419	0.339	0.818	FALSE
BSV	7	30	2.230	1.460	7.580	FALSE
SNO	1	3	-0.470	-0.470	-0.470	TRUE
Total	665	4487	0.521	0.292	0.936	FALSE